

## Problem

Huntington's disease (HD) is a terminal and usually leads to death within 15 to 20 years after the onset of the disease (Kumar and Ratan, 2016). It affects around 9 of every 100,000 people (Medina et al., 2022). Current treatments also have side affects and do not target the oxidative stress.



<https://inyolasvegas.com/blog/geraniol/>  
Figure 1. Geraniol and palmarosa flower

## Background

HD is a neurodegenerative disease caused by a CAG expansion in the Htt gene, (Siddique et al., 2021). Patients with CAG repeat lengths of 36 or higher in the Htt gene are considered to be at high risk for the disease.

| Huntington's status                | CAG repeat length                                                                       |
|------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Unaffected</b> Normal           |  10-26 |
| Intermediate allele                |  27-35 |
| <b>Affected</b> Reduced penetrance |  36-39 |
| Full penetrance                    |  40+   |

Figure 2. This image shows the difference in CAG repeats between diseased and normal Htt. Image is from National Institute of Standards and Technology.

- **Motor symptoms** typically arise in the form of **Chorea**, defined by abnormal involuntary jerks, dystonia, or tics (Ferguson et al., 2022).
- **CAG repeat length** has a large effect on HD onset and **severity**, with higher repeat lengths leading to more cognitive decline (Langbehn et al., 2019).
- Oxidative stress (OS) is often associated with Huntington's disease, and is **higher** in HD patients (Kumar and Ratan, 2016).
- An increase in OS in the cells can cause permanent damage to biomolecules and changes in signaling pathways, leading to more neurodegeneration.

**Geraniol**, a monoterpene alcohol, was used as a potential treatment. It has **antioxidant** and **anti-neurodegenerative properties** (Bagheri et al., 2023). It has worked in reducing motor dysfunction OS in other neurodegenerative diseases, showing potential to work on HD.

## Research Question

What is the effect of geraniol on motor dysfunction in *D. melanogaster*, and does its efficiency change based on CAG repeat length of the Htt gene of HD *D. melanogaster*?

## Hypothesis

A higher concentration of Geraniol would lead to lower levels of oxidative stress in the tissues of the different HD *D. melanogaster* stocks and increased motor ability. Also, the *D. melanogaster* model with higher repeat lengths in the Htt gene will show smaller improvements in motor ability when compared to those with lower repeat lengths.

# Determining the Effect of the Antioxidant Geraniol on the Motor Dysfunction and Oxidative Stress in the *Drosophila Melanogaster* Model of Huntington's Disease

## Methodology

**Independent variables:** CAG repeat length (96QHD or 72QHD) and geraniol concentration (0mM or 40mM).  
**Dependent variable:** motor dysfunction quantified through the climbing assay, and oxidative stress, quantified through the TBARS assay. More motor dysfunction is linked to lower climbing ability and higher oxidative stress is represented by higher TBARS concentrations

groups:

- Gal4 + 0mM geraniol- control
- Gal4 +40mM geraniol
- 72Q HD +0 mM geraniol-negative control
- 72Q HD +40mM geraniol
- 96Q HD+ 0mM geraniol-negative control
- 96Q HD + 40mM geraniol.

### TBARS Assay Protocol

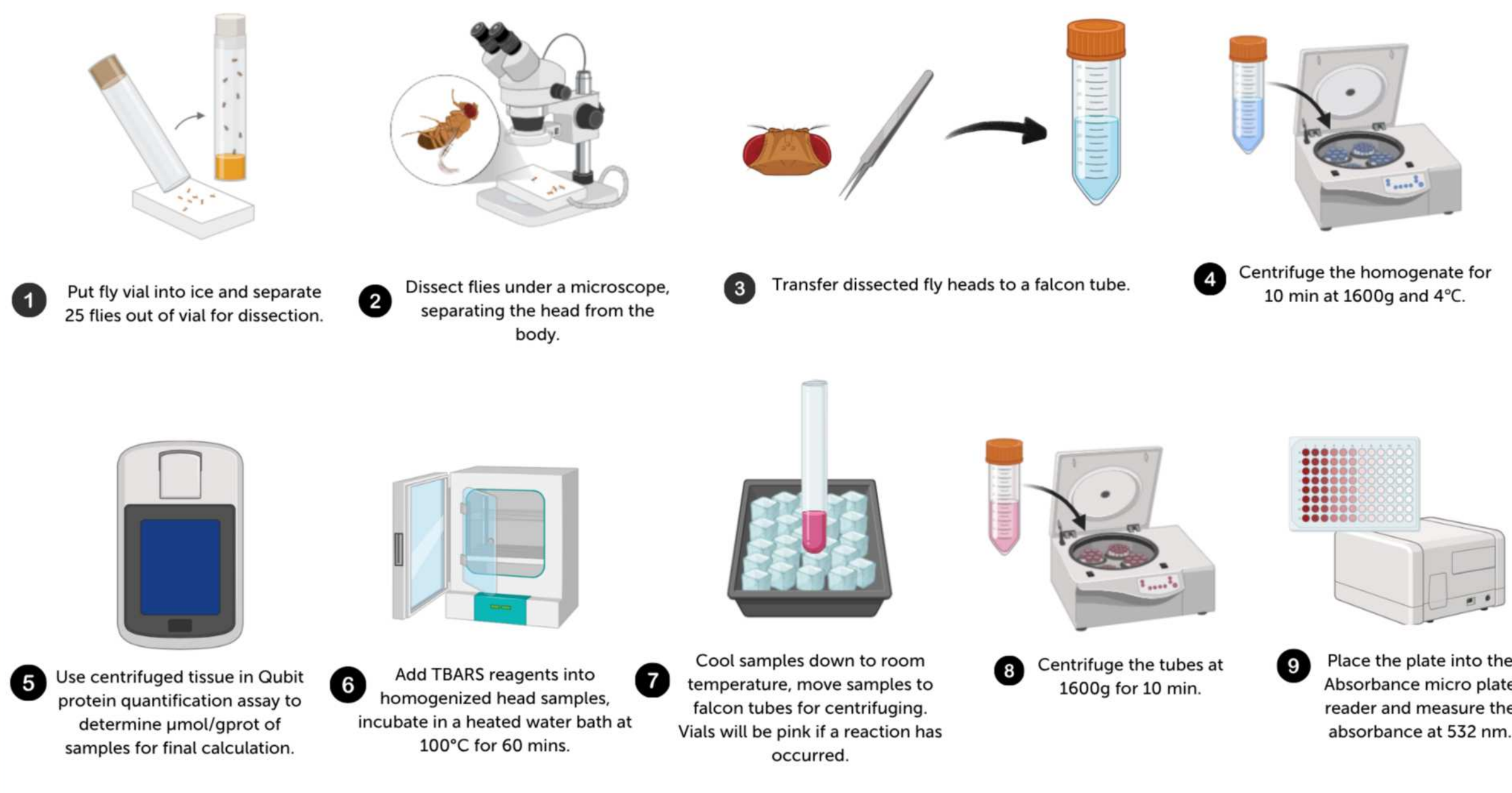


Figure 3. TBARS assay procedure.

### Climbing Assay Protocol

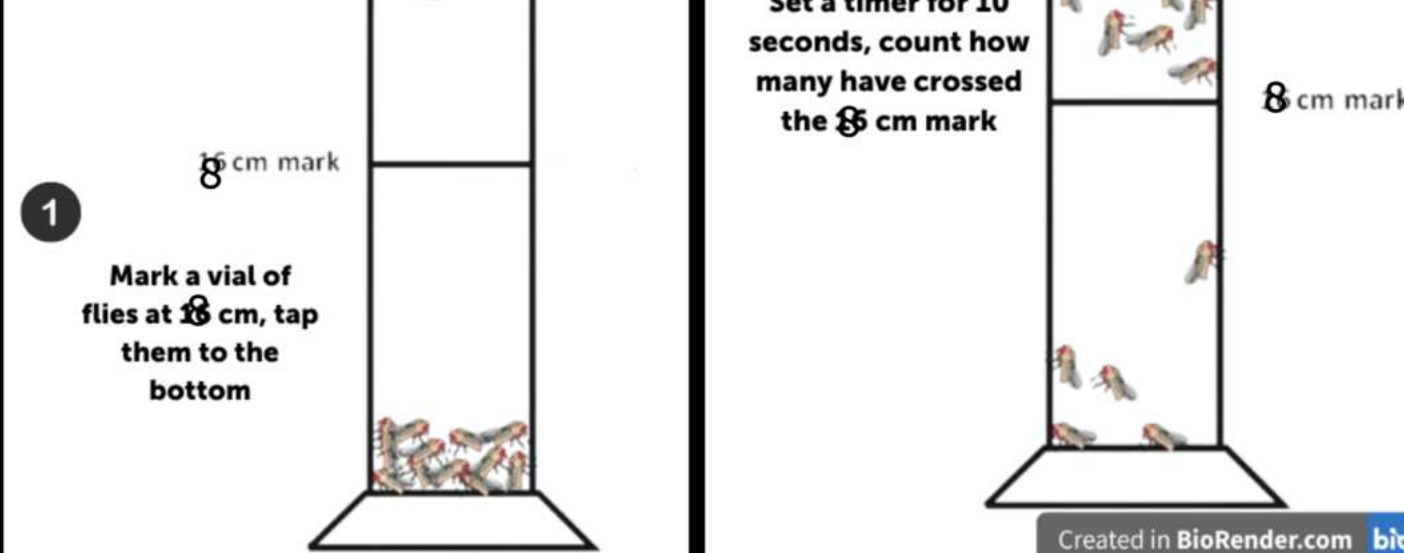


Figure 4. Climbing assay procedure.



Figure 5. Picture of flies used.

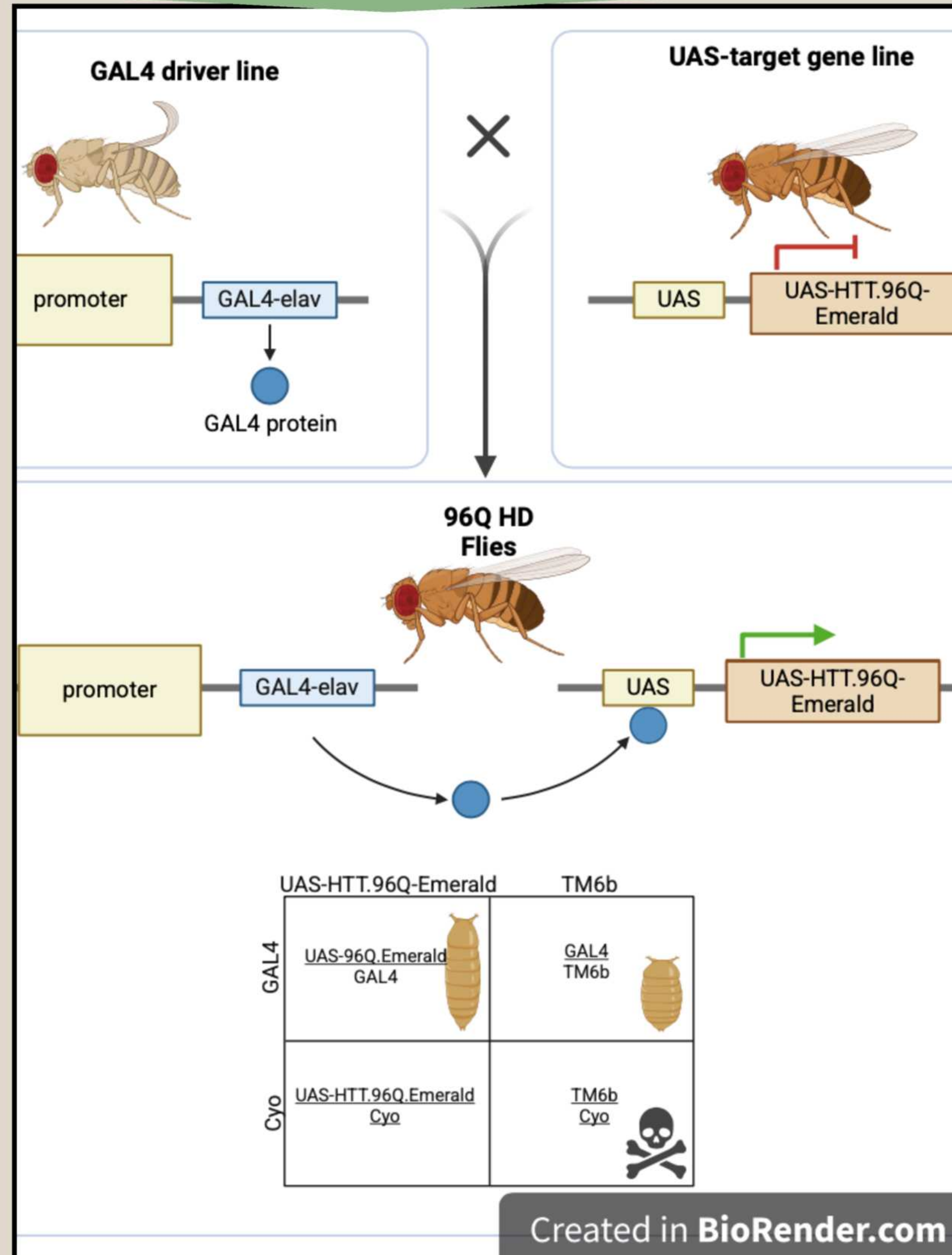


Figure 6. The figure above shows the crossing matrix used for the 72Q and 96Q HD crosses

## Results

An ANOVA test was run on the climbing assay data collected for each of the 6 groups.

**Null Hypothesis: No difference in climbing ability between groups.**

**Alternate hypothesis: Not all the groups are the same.**

The statistical significance value was set at 0.05. The test yielded a p value of <0.01.

An Kruskal Wallis test was run on the TBARS assay data collected for each of the 6 groups.

**Null Hypothesis: No difference in TBARS between groups.**

**Alternate hypothesis: Not all the groups are the same.**

The statistical significance value was set at 0.05. The test yielded a p value of 0.09.

Figure 7. The figure above shows the TBARS assay results for the groups

\*All tables and figures were created by the researcher unless stated otherwise

## Results

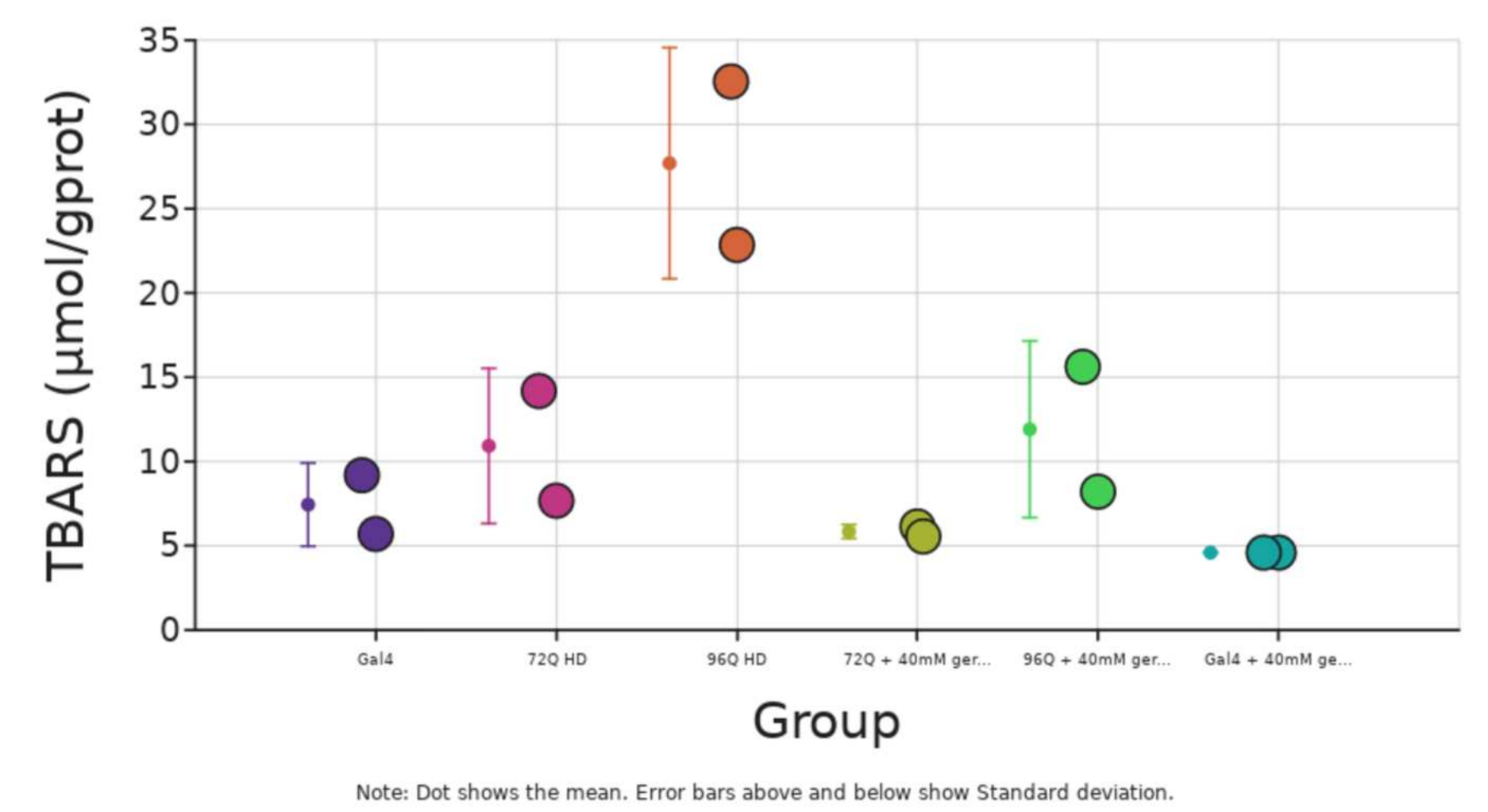


Figure 8 , comparison of TBARS values between groups

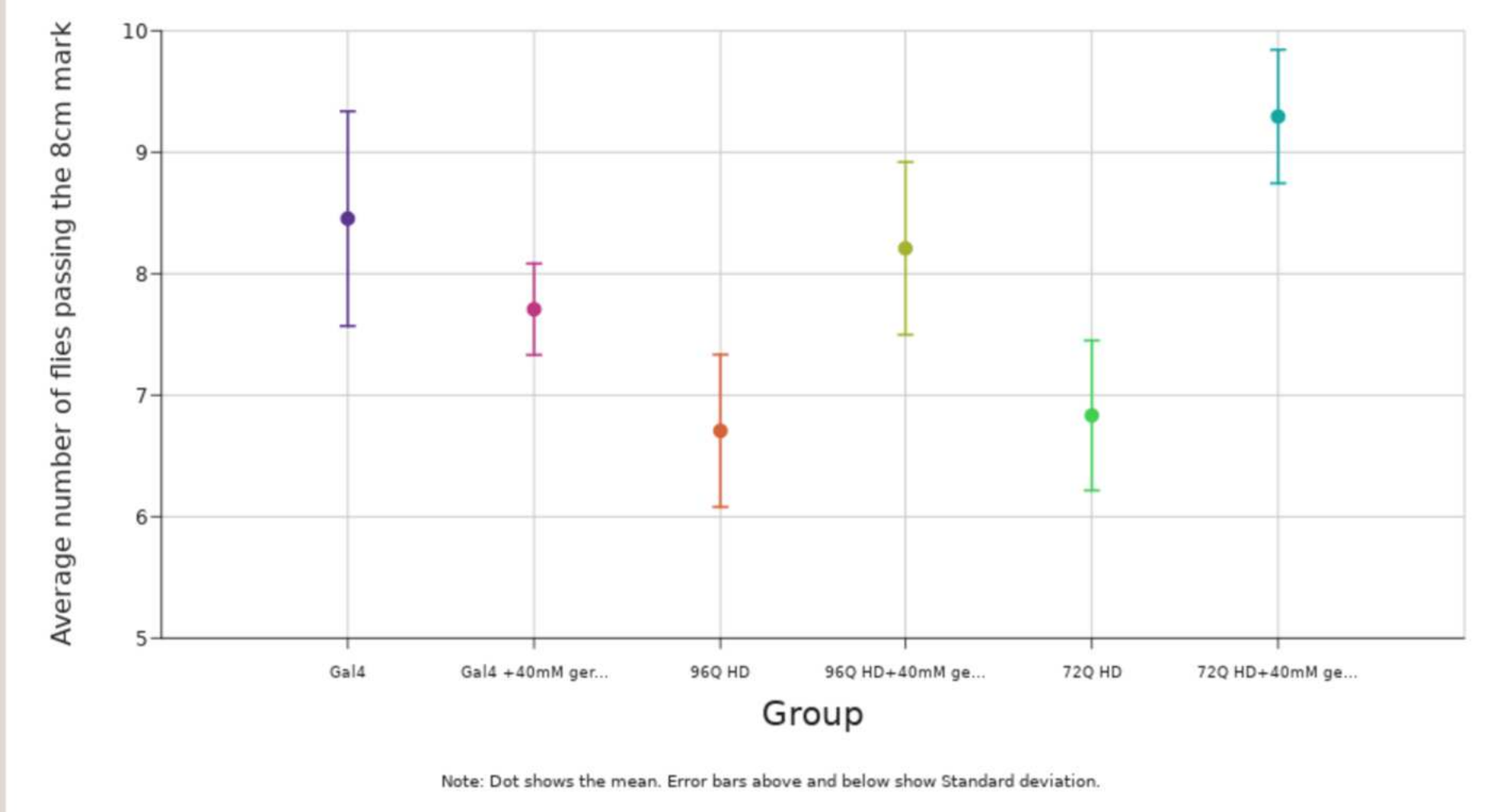


Figure 9, comparison of average flies passing 8cm mark.

## Discussion

- Statistically significant differences in the 72Q HD and 96QHD controls compared to experimental groups supports the hypothesis that geraniol treatment can help relieve motor dysfunction, and thereby indicate a possible reduction in oxidative stress.
- Although no difference was seen between the 96QHD+0mM and 72QHD+0mM groups at baseline, treatment of geraniol did result in different levels of improvement.
- Indicates that geraniol may have a different efficiency based on CAG repeat length, validating the hypothesis.
- No statistically significant differences in TBARS were observed in either the 72Q HD+40mM or the 96Q HD+40mM groups when compared to the 72Q HD and 96Q HD groups. However, the average TBARS was lower for both groups after treatment, indicating the possibility of a reduction

This work shows the potential for geraniol to reduce the motor dysfunction associated with Huntington's disease.

- Future research should focus on testing different concentrations of geraniol
- Due to time limitations, we did not get many TBARS assay trials conducted for the different groups, which should be a focus of future research.
- Geraniol may be a future therapeutic option for reducing these symptoms in HD patients.

## References

- Bagheri, S., Rashno, M., Salehi, I., Karimi, S. A., Raoufi, S., & Komaki, A. (2023). Geraniol improves passive avoidance memory and hippocampal synaptic plasticity deficits in a rat model of Alzheimer's disease. *European Journal of Pharmacology*, 951, 1397–1408. <https://doi.org/10.1016/j.ejphar.2023.175714>
- Bono-Yagüe, J., Gómez-Escribano, A. P., Millán, J. M., & Vázquez-Manrique, R. P. (2020). Reactive species in Huntington disease: Are they really the radicals you want to catch?. *Antioxidants*, 9(7), 577. <https://doi.org/10.3390/antiox9070577>
- Ferguson MW, Kennedy CJ, Palpagama TH, Waldvogel HJ, Fauli RLM, Kwakowsky A. (2022). Current and possible future therapeutic options for Huntington's disease. *Journal of Central Nervous System Disease*. 14. <https://doi.org/10.1177/11795735221092517>
- Kumar, A., & Ratan, R. R. (2016). Oxidative stress and Huntington's disease: The good, the bad, and the ugly. *Journal of Huntington's Disease*, 5(3), 217–237. <https://doi.org/10.3233/JHD-160205>
- Langbehn, D. R., Stout, J. C., Gregory, S., Mills, J. A., Durr, A., Leavitt, B. R., Roos, R. A. C., Long, J. D., Owen, G., Johnson, H. J., Borowsky, B., Craufurd, D., Reilmann, R., Landwehrmeyer, G. B., Schall, R. I., Tabrizi, S. J., & TRACK-HD and Track-On HD Groups (2019). Association of CAG repeats with long-term progression in Huntington disease. *JAMA neurology*, 76(11), 1375–1385. <https://doi.org/10.1001/jamaneurol.2019.2368>